Source coding

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Codes

Definition 0.1 (Binary code)**.** Let X be a discrete random variable. A **binary code** for X assigns each value x a *codeword*, a finite string of binary digits, of length $\ell(x)$.

Definition 0.2 (Uniquely decodable code)**.** A binary code is **uniquely decodable** if every finite sequence of codewords, written one after another, can be split back into codewords in only one way.

Definition 0.3 (Expected length)**.** The **expected length** of a binary code for X, where X has probability mass function $p(x) \stackrel{\text{def}}{=} \mathbb{P}(X = x)$, is

$$L \stackrel{\text{def}}{=} \mathbb{E}[\ell(X)] = \sum_x p(x)\,\ell(x).$$

Remark 0.1 (Short codewords for common values)*.* A code with a small expected length gives frequent values short codewords and rare values long ones, since each value's length counts in L in proportion to its probability.

The source coding theorem

Theorem 0.1 (Source coding theorem)**.** *Let X be a discrete random variable that takes finitely many values, at least two of them with positive probability, and let $H_2(X)$ be its entropy with base-2 logarithms.*

1. *Every uniquely decodable binary code for X has expected length $L \geq H_2(X)$.*

2. *Some uniquely decodable binary code for X has expected length $L < H_2(X) + 1$.*

3. *Coding blocks of n independent copies of X at a time, some uniquely decodable binary code spends, on average, fewer than $H_2(X) + 1/n$ binary digits per copy.*

Remark 0.2 (Where the theorem comes from). Shannon (1948) proved that entropy is the limit on how far the output of a source can be compressed. Cover and Thomas (2006, sec. 5.3 to 5.5) prove each part of Theorem 0.1. For part 2, they give each value x with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 p(x) \rceil$, its surprisal in bits rounded up; the Kraft inequality (Cover and Thomas 2006, sec. 5.2) guarantees that a code with these lengths exists.

Coding with the wrong distribution

Theorem 0.2 (The cost of coding with the wrong distribution)**.** *Let X take finitely many values, with probability mass function p, and let q be a probability mass function, possibly different from p, with $q(x) > 0$ wherever $p(x) > 0$. Suppose a code gives each value x with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 q(x) \rceil$. Then its expected length L under p satisfies*

$$H_2(p, q) \leq L < H_2(p, q) + 1,$$

where $H_2(p, q)$ is the cross-entropy of q relative to p, with base-2 logarithms.

Proof

Proof. Every sum here runs over the values x with $p(x) > 0$. Rounding up adds less than one binary digit, so for each such x

$$-\log_2 q(x) \leq \ell(x) < -\log_2 q(x) + 1.$$

Multiply each part by $p(x) > 0$ and sum. Since $\sum_x p(x) = 1$,

$$-\sum_x p(x) \log_2 q(x) \leq \sum_x p(x)\, \ell(x) < -\sum_x p(x) \log_2 q(x) + 1.$$

The middle sum is L, and the outer sums are $H_2(p, q)$. □

Theorem 0.3 (Cross-entropy is entropy plus divergence)**.** *Let p and q be probability mass functions on the values $x_1, \ldots, x_n$, with $q(x) > 0$ wherever $p(x) > 0$. Write $H_2(p)$ for the entropy of p, $H_2(p, q)$ for the cross-entropy of q relative to p, and $D_2(p \,\|\, q)$ for the Kullback–Leibler divergence of q from p, all with base-2 logarithms. Then*

$$H_2(p, q) = H_2(p) + D_2(p \,\|\, q).$$

Proof

Proof. Every sum here runs over the values x with $p(x) > 0$. For every such x,

$$-\log_2 q(x) = -\log_2 p(x) + \log_2 \frac{p(x)}{q(x)}.$$

Multiply by $p(x)$ and add over x:

$$-\sum_x p(x) \log_2 q(x) = -\sum_x p(x) \log_2 p(x) + \sum_x p(x) \log_2 \frac{p(x)}{q(x)}.$$

Reading left to right, these are $H_2(p, q)$, $H_2(p)$ and $D_2(p \,\|\, q)$. □

Example 0.1 (Cross-entropy, entropy and divergence for a uniform code)**.** Let $p = (1/2, 1/4, 1/4)$ be the probability mass function of a three-valued random variable, and let $q = (1/3, 1/3, 1/3)$ assign $1/3$ to each of the three values. The code below computes the cross-entropy, the entropy and the divergence in bits.

```
p <- c(1 / 2, 1 / 4, 1 / 4)
q <- rep(1 / 3, 3)
cross_entropy <- -sum(p * log2(q))
shannon_entropy <- -sum(p * log2(p))
divergence <- sum(p * log2(p / q))
c(
  cross_entropy = cross_entropy,
  shannon_entropy = shannon_entropy,
  divergence = divergence
)
#>   cross_entropy shannon_entropy      divergence
#>       1.5849625       1.5000000       0.0849625
```

The cross-entropy is $H_2(p, q) = 1.585$ bits. The entropy is $H_2(p) = 1.5$ bits, and the divergence is $D_2(p \,\|\, q) = 0.085$ bits. They satisfy Theorem 0.3: $1.5 + 0.085 = 1.585$.

Theorem 0.4 (The divergence is nonnegative)**.** *Let p and q be probability mass functions on the values $x_1, \ldots, x_n$, with $q(x) > 0$ wherever $p(x) > 0$. Then the Kullback–Leibler divergence satisfies*

$$D_2(p \,\|\, q) \geq 0,$$

with $D_2(p \,\|\, q) = 0$ *exactly when* $q = p$.

Proof

Proof. Every sum here runs over the values x with $p(x) > 0$. For every $t > 0$, $\ln t \leq t - 1$, with equality exactly when $t = 1$. Apply this with $t = q(x)/p(x)$, and multiply by $p(x) > 0$:

$$p(x) \ln \frac{q(x)}{p(x)} \leq q(x) - p(x),$$

with equality exactly when $q(x) = p(x)$. Add over x. Since $\sum_x p(x) = 1$ and $\sum_x q(x) \leq 1$,

$$\sum_x p(x) \ln \frac{q(x)}{p(x)} \leq \sum_x q(x) - 1 \leq 0.$$

The left side is $-\ln 2 \cdot D_2(p \,\|\, q)$, so $D_2(p \,\|\, q) \geq 0$.
If $D_2(p \,\|\, q) = 0$, both relations hold with equality. The first forces $q(x) = p(x)$ at every x with $p(x) > 0$. The second then gives $\sum_x q(x) = 1$, so q has no mass where p has none, and $q = p$. Conversely, $q = p$ gives $\log_2(p(x)/q(x)) = 0$ for every x. □

Example 0.2 (Divergence of a biased coin from a fair coin)**.**
Let $p = (0.9, 0.1)$ be the probability mass function of a biased coin, and let $q = (0.5, 0.5)$ be a fair coin. The code below computes the divergence both ways, in bits.

```
p <- c(0.9, 0.1)
q <- c(0.5, 0.5)
c(
  p_from_q = sum(p * log2(p / q)),
  q_from_p = sum(q * log2(q / p)),
  p_from_p = sum(p * log2(p / p))
)
#>  p_from_q  q_from_p  p_from_p
#> 0.5310044 0.7369656 0.0000000
```

Both divergences between the two coins are positive, as Theorem 0.4 requires, and they differ, so swapping p and q changes the divergence. The divergence of p from itself is 0.

Remark 0.3 (Cross-entropy is the cost of the wrong code). Theorem 0.3 splits the cross-entropy into the entropy and the divergence. By Theorem 0.1, no uniquely decodable code averages fewer than $H_2(p)$ bits. By Theorem 0.2, a code built for q costs about $D_2(p \,\|\, q)$ bits per value more than that. By Theorem 0.4, the excess vanishes only when the code was built for p (Cover and Thomas 2006, sec. 2.6; Yeung 2008, sec. 2.5). Cover and Thomas (2006, sec. 5.4) state Theorem 0.2 in this form, with $H_2(p) + D_2(p \,\|\, q)$ in place of $H_2(p, q)$.

Remark 0.4 (Why models are fit by minimizing cross-entropy). Fitting a classifier by minimizing cross-entropy chooses the model's distribution q so that a code built for q would spend as few bits as possible on the observed labels. By Remark 0.3, that pushes q toward the distribution the labels came from.
For a single labeled example, let e be the distribution that puts all its mass on the observed label. Then $H_2(e) = 0$ (as for a coin that always lands heads), so the example's cross-entropy loss $H_2(e, q)$ equals $D_2(e \,\|\, q)$: every bit the loss reports is a wasted one.

References

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